CHAPTER 3
WORKING WITH FINANCIAL STATEMENTS

Answers to Concepts Review and Critical Thinking Questions

1. *a.* If inventory is purchased with cash, then there is no change in the current ratio. If inventory is purchased on credit, then there is a decrease in the current ratio if it was initially greater than 1.0.

b. Reducing accounts payable with cash increases the current ratio if it was initially greater than 1.0

c. Reducing short-term debt with cash increases the current ratio if it was initially greater than 1.0.

d. As long-term debt approaches maturity, the principal repayment and the remaining interest expense become current liabilities. Thus, if debt is paid off with cash, the current ratio increases if it was initially greater than 1.0. If the debt has not yet become a current liability, then paying it off will reduce the current ratio since current liabilities are unaffected.

e. Reduction of accounts receivables and an increase in cash leaves the current ratio unchanged.

f. Inventory sold at cost reduces inventory and raises cash, so the current ratio is unchanged.

g. Inventory sold for a profit raises cash in excess of the inventory recorded at cost, so the current ratio increases.

2. The firm has increased inventory relative to other current assets; therefore, assuming current liability levels remain mostly unchanged, liquidity has potentially decreased.

3. A current ratio of .50 means that the firm has twice as much in current liabilities as it does in current assets; the firm potentially has poor liquidity. If pressed by its short-term creditors and suppliers for immediate payment, the firm might have a difficult time meeting its obligations. A current ratio of 1.50 means the firm has 50% more current assets than it does current liabilities. This probably represents an improvement in liquidity; short-term obligations can generally be met completely with a safety factor built in. A current ratio of 15.0, however, might be excessive. Any excess funds sitting in current assets generally earn little or no return. These excess funds might be put to better use by investing in productive long-term assets or distributing the funds to shareholders.

4. *a.* Quick ratio provides a measure of the short-term liquidity of the firm, after removing the effects of inventory, generally the least liquid of the firm's current assets.

b. Cash ratio represents the ability of the firm to completely pay off its current liabilities balance with its most liquid asset (cash).

c. The capital intensity ratio tells us the dollar amount investment in assets needed to generate one dollar in sales.

d. Total asset turnover measures how much in sales is generated by each dollar of firm assets.

e. Equity multiplier represents the degree of leverage for an equity investor of the firm; it measures the dollar worth of firm assets each equity dollar has a claim to.

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f. Times interest earned ratio provides a relative measure of how well the firm's operating earnings can cover current interest obligations.

g. Profit margin is the accounting measure of bottom-line profit per dollar of sales.

h. Return on assets is a measure of bottom-line profit per dollar of total assets.

i. Return on equity is a measure of bottom-line profit per dollar of equity.

j. Price-earnings ratio reflects how much value per share the market places on a dollar of accounting earnings for a firm.

5. Common-size financial statements express all balance sheet accounts as a percentage of total assets and all income statement accounts as a percentage of total sales. Using these percentage values rather than nominal dollar values facilitates comparisons between firms of different sizes or business types.

6. Peer group analysis involves comparing the financial ratios and operating performance of a particular firm to a set of peer group firms in the same industry or line of business. Comparing a firm to its peers allows the financial manager to evaluate whether some aspects of the firm's operations, finances, or investment activities are out of line with the norm, thereby providing some guidance on appropriate actions to take to adjust these ratios, if appropriate. An aspirant group would be a set of firms whose performance the company in question would like to emulate. The financial manager often uses the financial ratios of aspirant groups as the target ratios for his or her firm; some managers are evaluated by how well they match the performance of an identified aspirant group.

7. Return on equity is probably the most important accounting ratio that measures the bottom-line performance of the firm with respect to the equity shareholders. The DuPont identity emphasizes the role of a firm's profitability, asset utilization efficiency, and financial leverage in achieving a ROE figure. For example, a firm with ROE of 20% would seem to be doing well, but this figure may be misleading if it were a marginally profitable (low profit margin) and highly levered (high equity multiplier). If the firm's margins were to erode slightly, the ROE would be heavily impacted.

8. The book-to-bill ratio is intended to measure whether demand is growing or falling. It is closely followed because it is a barometer for the entire high-tech industry where levels of revenues and earnings have been relatively volatile.

9. If a company is growing by opening new stores, then presumably total revenues would be rising. Comparing total sales at two different points in time might be misleading. Same-store sales control for this by only looking at revenues of stores open within a specific period.

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10. *a.* For an electric utility such as Con Ed, expressing costs on a per kilowatt-hour basis would be a way of comparing costs with other utilities of different sizes.

b. For a retailer such as JC Penney, expressing sales on a per square foot basis would be useful in comparing revenue production against other retailers.

c. For an airline such as Delta, expressing costs on a per passenger mile basis allows for comparisons with other airlines by examining how much it costs to fly one passenger one mile.

d. For an online service such as Google or Yahoo!, using a per web hit basis for costs would allow for comparisons with similar services.

e. For a hospital such as Holy Cross, revenues and costs expressed on a per bed basis would be useful.

f. For a college textbook publisher such as McGraw-Hill Higher Education, the leading publisher of finance textbooks for the college market, the obvious standardization would be per book sold.

11. As with any ratio analysis, the ratios themselves do not necessarily indicate a problem, but indicate that something is different, and it is up to us to determine if a problem exists. If the cost of goods sold as a percentage of sales is increasing, we would expect that EBIT as a percentage of sales would decrease, all else constant. An increase in the cost of goods sold as a percentage of sales occurs because the cost of raw materials or other inventory is increasing at a faster rate than the sales price.

This may be a bad sign since the contribution of each sales dollar to net income and cash flow is lower. However, when a new product, for example, the HDTV, enters the market, the price of one unit will often be high relative to the cost of goods sold per unit, and demand, therefore sales, will initially be small. As the product market becomes more developed, price of the product generally drops, and sales increase as more competition enters the market. In this case, the increase in cost of goods sold as a percentage of sales is to be expected. The maker or seller expects to boost sales at a faster rate than its cost of goods sold increases. In this case, a good practice would be to examine the common-size income statements to see if this is an industry-wide occurrence.

12. If we assume that the cause is negative, the two reasons for the trend of increasing cost of goods sold as a percentage of sales are that costs are becoming too high or the sales price is not increasing fast enough. If the cause is an increase in the cost of goods sold, the manager should look at possible actions to control costs. If costs can be lowered by seeking lower cost suppliers of similar or higher quality, the cost of goods sold as a percentage of sales should decrease. Another alternative is to increase the sales price to cover the increase in the cost of goods sold. Depending on the industry, this may be difficult or impossible. For example, if the company sells most of its products under a long-term contract that has a fixed price, it may not be able to increase the sales price and will be forced to look for other cost-cutting possibilities. Additionally, if the market is competitive, the company might also be unable to increase the sales price.

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Solutions to Questions and Problems

NOTE: All end-of-chapter problems were solved using a spreadsheet. Many problems require multiple steps. Due to space and readability constraints, when these intermediate steps are included in this solutions manual, rounding may appear to have occurred. However, the final answer for each problem is found without rounding during any step in the problem.

Basic

1. Using the formula for NWC, we get:

NWC = CA – CL
CA = CL + NWC
CA = $5,730 + 2,945
CA = $8,675

So, the current ratio is:

Current ratio = CA/CL
Current ratio = $8,675/$5,730
Current ratio = 1.51 times

And the quick ratio is:

Quick ratio = (CA – Inventory)/CL
Quick ratio = ($8,675 – 3,720)/$5,730
Quick ratio = .86 times

2. We need to find net income first. So:

Profit margin = Net income/Sales
Net income = Profit margin(Sales)
Net income = .05($13,400,000)
Net income = $670,000

ROA = Net income/TA
ROA = $670,000/$10,200,000
ROA = .0657, or 6.57%

To find ROE, we need to find total equity. Since TL & OE equals TA:

TA = TD + TE
TE = TA – TD
TE = $10,200,000 – 4,900,000
TE = $5,300,000

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ROE = Net income/TE
ROE = $670,000/$5,300,000
ROE = .1264, or 12.64%

3. Receivables turnover = Sales/Receivables

Receivables turnover = $4,986,340/$473,380
Receivables turnover = 10.53 times

Days' sales in receivables = 365 days/Receivables turnover
Days' sales in receivables = 365/10.53
Days' sales in receivables = 34.65 days

On average, the company's customers paid off their accounts in 34.65 days.

4. Inventory turnover = COGS/Inventory

Inventory turnover = $4,738,216/$385,342
Inventory turnover = 12.30 times

Days' sales in inventory = 365 days/Inventory turnover
Days' sales in inventory = 365/12.30
Days' sales in inventory = 29.68 days

On average, a unit of inventory sat on the shelf 29.68 days before it was sold.

5. Total debt ratio = .44 = TD/TA

Substituting total debt plus total equity for total assets, we get:

.44 = TD/(TD + TE)

Solving this equation yields:

.44(TE) = .56(TD)

Debt-equity ratio = TD/TE = .44/.56 = .79

Equity multiplier = 1 + D/E = 1.79

6. Net income = Addition to RE + Dividends = $395,000 + 245,000 = $640,000

Earnings per share = NI/Shares = $640,000/165,000 = $3.88 per share

Dividends per share = Dividends/Shares = $245,000/165,000 = $1.48 per share

Book value per share = TE/Shares = $5,700,000/165,000 = $34.55 per share

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Market-to-book ratio	= Share price/BVPS	= \$78/\$34.55 = 2.26 times
PE ratio	= Share price/EPS	= \$78/\$3.88 = 20.11 times
Sales per share	= Sales/Shares	= \$7,450,000/165,000 = \$45.15
P/S ratio	= Share price/Sales per share	= \$78/\$45.15 = 1.73 times

7. ROE = (PM)(TAT)(EM)
ROE = (.0605)(1.64)(1.27)
ROE = .1260, or 12.60%

8. This question gives all of the necessary ratios for the DuPont identity except the equity multiplier, so, using the DuPont identity:

ROE = (PM)(TAT)(EM)
ROE = .1385 = (.054)(1.57)(EM)

EM = .1385/(.054)(1.57)
EM = 1.63

D/E = EM – 1
D/E = 1.63 – 1
D/E = .63

9. To find the days' sales in payables, we first need to find the payables turnover. The payables turnover was:

Payables turnover = COGS/Accounts payable
Payables turnover = \$83,261/\$18,452
Payables turnover = 4.51 times

Now, we can use the payables turnover to find the days' sales in payables as:

Days' sales in payables = 365 days/Payables turnover
Days' sales in payables = 365/4.51
Days' sales in payables = 80.89 days

The company left its bills to suppliers outstanding for 80.89 days, on average. A large value for this ratio could imply that either (1) the company is having liquidity problems, making it difficult to pay off its short-term obligations, or (2) that the company has successfully negotiated lenient credit terms from its suppliers.

10. The equity multiplier is:

EM = 1 + D/E
EM = 1 + .73
EM = 1.73

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One formula to calculate return on equity is:

ROE = ROA(EM)
ROE = .079(1.73)
ROE = .1367, or 13.67%

ROE can also be calculated as:

ROE = NI/TE

So, net income is:

Net income = ROE(TE)
Net income = .1367($645,000)
Net income = $88,152.15

11. To find the internal growth rate, we need the plowback, or retention, ratio. The plowback ratio is:

$b = 1 - .35$
$b = .65$

Now, we can use the internal growth rate equation to find:

Internal growth rate = [(ROA)(*b*)]/[1 – (ROA)(*b*)]
Internal growth rate = [.076(.65)]/[1 – .076(.65)]
Internal growth rate = .0520, or 5.20%

12. To find the sustainable growth rate we need the plowback, or retention, ratio. The plowback ratio is:

$b = 1 - .20$
$b = .80$

Now, we can use the sustainable growth rate equation to find:

Sustainable growth rate = [(ROE)(*b*)]/[1 – (ROE)(*b*)]
Sustainable growth rate = [.141(.80)]/[1 – .141(.80)]
Sustainable growth rate = .1271, or 12.71%

13. We need the return on equity to calculate the sustainable growth rate. To calculate return on equity, we need to realize that the total asset turnover is the inverse of the capital intensity ratio and the equity multiplier is one plus the debt-equity ratio. So, the return on equity is:

ROE = (Profit margin)(Total asset turnover)(Equity multiplier)
ROE = (.081)(1/.95)(1 + 1.05)
ROE = .1748, or 17.48%

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Next, we need the plowback ratio. The plowback ratio is one minus the payout ratio. We can calculate the payout ratio as the dividends divided by net income, so the plowback ratio is:

$b = 1 – \$24,000/\$97,000$
$b = .75$

Now we can use the sustainable growth rate equation to find:

Sustainable growth rate = [(ROE)(*b*)]/[1 – (ROE)(*b*)]
Sustainable growth rate = [.1748(.75)]/[1 – .1748(.75)]
Sustainable growth rate = .1515, or 15.15%

14. We need the return on equity to calculate the sustainable growth rate. Using the DuPont identity, the return on equity is:

ROE = (Profit margin)(Total asset turnover)(Equity multiplier)
ROE = (.052)(2.90)(.95)
ROE = .1433, or 14.33%

To find the sustainable growth rate, we need the plowback, or retention, ratio. The plowback ratio is:

$b = 1 – .35$
$b = .65$

Now, we can use the sustainable growth rate equation to find:

Sustainable growth rate = [(ROE)(*b*)]/[1 – (ROE)(*b*)]
Sustainable growth rate = [.1433(.65)]/[1 – .1433(.65)]
Sustainable growth rate = .1027, or 10.27%

15. To calculate the common-size balance sheet, we divide each asset account by total assets, and each liability and equity account by total liabilities and equity. For example, the common-size cash percentage for 2024 is:

Cash percentage = Cash/Total assets
Cash percentage = \$30,290/\$1,215,582
Cash percentage = .0249, or 2.49%

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Repeating this procedure for each account, we get:

	2024		2025	
Assets				
Current assets				
Cash	$30,290	2.49%	$34,521	2.52%
Accounts receivable	72,981	6.00%	85,708	6.26%
Inventory	172,442	14.19%	204,086	14.90%
Total	$275,713	22.68%	$324,315	23.68%
Fixed assets				
Net plant and equipment	$939,869	77.32%	$1,045,107	76.32%
Total assets	$1,215,582	100%	$1,369,422	100%
Liabilities and owners' equity				
Current liabilities				
Accounts payable	$217,931	17.93%	$230,828	16.86%
Notes payable	100,645	8.28%	115,491	8.43%
Total	$318,576	26.21%	$346,319	25.29%
Long-term debt	$379,000	31.18%	$401,000	29.28%
Owners' equity				
Common stock and paid-in surplus	$192,000	15.79%	$201,000	14.68%
Accumulated retained earnings	326,006	26.82%	421,103	30.75%
Total	$518,006	42.61%	$622,103	45.43%
Total liabilities and owners' equity	$1,215,582	100%	$1,369,422	100%

16. *a.* The current ratio is calculated as:

Curent ratio = Current assets/Current liabilities

Current ratio$_{2024}$ = $275,713/$318,576
Current ratio$_{2024}$ = .87 times

Current ratio$_{2025}$ = $324,315/$346,319
Current ratio$_{2025}$ = .94 times

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b. The quick ratio is calculated as:

Quick ratio = (Current assets – Inventory)/Current liabilities

Quick ratio$_{2024}$ = ($275,713 – 172,442)/$318,576
Quick ratio$_{2024}$ = .32 times

Quick ratio$_{2025}$ = ($324,315 – 204,086)/$346,319
Quick ratio$_{2025}$ = .35 times

c. The cash ratio is calculated as:

Cash ratio = Cash/Current liabilities

Cash ratio$_{2024}$ = $30,290/$318,576
Cash ratio$_{2024}$ = .10 times

Cash ratio$_{2025}$ = $34,521/$346,319
Cash ratio$_{2025}$ = .10 times

d. The debt-equity ratio is calculated as:

Debt-equity ratio = Total debt/Total equity
Debt-equity ratio = (Current liabilities + Long-term debt)/Total equity

Debt-equity ratio$_{2024}$ = ($318,576 + 379,000)/$518,006
Debt-equity ratio$_{2024}$ = 1.35 times

Debt-equity ratio$_{2025}$ = ($346,319 + 401,000)/$622,103
Debt-equity ratio$_{2025}$ = 1.20 times

And the equity multiplier is:

Equity multiplier = 1 + Debt-equity ratio

Equity multiplier$_{2024}$ = 1 + 1.35
Equity multiplier$_{2024}$ = 2.35 times

Equity multiplier$_{2025}$ = 1 + 1.20
Equity multiplier$_{2025}$ = 2.20 times

e. The total debt ratio is calculated as:

Total debt ratio = Total debt/Total assets
Total debt ratio = (Current liabilities + Long-term debt)/Total assets

Total debt ratio$_{2024}$ = ($318,576 + 379,000)/$1,215,582
Total debt ratio$_{2024}$ = .57 times

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Total debt ratio$_{2025}$ = ($346,319 + 401,000)/$1,369,422
Total debt ratio$_{2025}$ = .55 times

17. Using the DuPont identity to calculate ROE, we get:

ROE = (Profit margin)(Total asset turnover)(Equity multiplier)
ROE = (Net income/Sales)(Sales/Total assets)(Total assets/Total equity)
ROE = ($86,432/$2,751,332)($2,751,332/$1,369,422)($1,369,422/$622,103)
ROE = .1389, or 13.89%

18. One equation to calculate ROA is:

ROA = (Profit margin)(Total asset turnover)

We can solve this equation to find total asset turnover as:

.058 = .051(Total asset turnover)
Total asset turnover = 1.14 times

Now, solve the ROE equation to find the equity multiplier, which is:

ROE = (ROA)(Equity multiplier)
.12 = .058(Equity multiplier)
Equity multiplier = 2.07 times

19. To calculate the ROA, we first need to find the net income. Using the profit margin equation, we find:

Profit margin = Net income/Sales
.057 = Net income/$12,700,000
Net income = $723,900

Now we can calculate ROA as:

ROA = Net income/Total assets
ROA = $723,900/$9,300,000
ROA = .0778, or 7.78%

20. To calculate the internal growth rate, we first need to calculate the ROA, which is:

ROA = NI/TA
ROA = $5,214/$49,500
ROA = .1053, or 10.53%

The plowback ratio, *b*, is one minus the payout ratio, so:

b = 1 – .30
b = .70

Now we can use the internal growth rate equation to get:

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Internal growth rate = (ROA × *b*)/[1 – (ROA × *b*)]
Internal growth rate = [.1053(.70)]/[1 – .1053(.70)]
Internal growth rate = .0796, or 7.96%

21. To calculate the sustainable growth rate, we first need to calculate the ROE, which is:

ROE = NI/TE
ROE = $5,214/$37,000
ROE = .1409, or 14.09%

The plowback ratio, *b*, is one minus the payout ratio, so:

b = 1 – .30
b = .70

Now we can use the sustainable growth rate equation to get:

Sustainable growth rate = (ROE × *b*)/[1 – (ROE × *b*)]
Sustainable growth rate = [.1409(.70)]/[1 – .1409(.70)]
Sustainable growth rate = .1094, or 10.94%

22. The total asset turnover is:

Total asset turnover = Sales/Total assets
Total asset turnover = $12,900,000/$6,800,000
Total asset turnover = 1.90 times

If the target total asset turnover is 2.10 times, we can use the total asset turnover equation to solve for the necessary sales level. The new sales level will be:

Total asset turnover = Sales/Total assets
2.10 = Sales/$6,800,000
Sales = $14,280,000

23. To find the ROE, we need the equity balance. Since we have the total debt, if we can find the total assets we can calculate the equity. Using the total debt ratio, we find total assets as:

Debt ratio = Total debt/Total assets
.65 = $674,000/Total assets
Total assets = $1,036,923

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Total liabilities and equity is equal to total assets. Using this relationship, we find:

Total assets = Total debt + Total equity
$1,036,923 = $674,000 + Total equity
Total equity = $362,923

Now, we can calculate the ROE as:

ROE = Net income/Total equity
ROE = $36,180/$362,923
ROE = .0997, or 9.97%

24. The earnings per share are:

EPS = Net income/Shares
EPS = $7,600,000/3,700,000
EPS = $2.05

The price-earnings ratio is:

PE = Price/EPS
PE = $38/$2.05
PE = 18.50 times

The sales per share are:

Sales per share = Sales/Shares
Sales per share = $46,300,000/3,700,000
Sales per share = $12.51

The price-sales ratio is:

P/S = Price/Sales per share
P/S = $38/$12.51
P/S = 3.04 times

The book value per share is:

Book value per share = Book value of equity/Shares
Book value per share = $21,700,000/3,700,000
Book value per share = $5.86 per share

And the market-to-book ratio is:

Market-to-book = Market value per share/Book value per share
Market-to-book = $38/$5.86
Market-to-book = 6.48 times

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25. To find the profit margin, we need the net income and sales. We can use the total asset turnover to find the sales and the return on assets to find the net income. Beginning with the total asset turnover, we find sales are:

Total asset turnover = Sales/Total assets
1.95 = Sales/$6,720,000
Sales = $13,104,000

And the net income is:

ROA = Net income/Total assets
.0790 = Net income/$6,720,000
Net income = $530,880

Now we can find the profit margin, which is:

Profit margin = Net income/Sales
Profit margin = $530,880/$13,104,000
Profit margin = .0405, or 4.05%

26. First, we need the enterprise value, which is:

Enterprise value = Market capitalization + Debt – Cash
Enterprise value = $875,000 + 240,000 – 43,000
Enterprise value = $1,072,000

And EBITDA is:

EBITDA = EBIT + Depreciation & Amortization
EBITDA = $98,160 + 173,000
EBITDA = $271,160

So, the enterprise value-EBITDA multiple is:

Enterprise value-EBITDA multiple = $1,072,000/$271,160
Enterprise value-EBITDA multiple = 3.95 times

Intermediate

27. This is a multistep problem involving several ratios. The ratios given are all part of the DuPont identity. The only DuPont identity ratio not given is the profit margin. If we know the profit margin, we can find the net income since sales are given. So, we begin with the DuPont identity:

ROE = .10 = (PM)(TAT)(EM) = (PM)(S/TA)(1 + D/E)

Solving the DuPont identity for profit margin, we get:

PM = [(ROE)(TA)]/[(1 + D/E)(S)]
PM = [(.10)($2,754)]/[(1 + .57)($6,785)]

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PM = .0259

Now that we have the profit margin, we can use this number and the given sales figure to solve for net income:

PM = .0259 = NI/S
NI = .0259($6,785)
Net income = $175.41

28. The solution to this problem requires a number of steps. First, remember that Current assets + Net fixed assets = Total assets. So, if we find the current assets and the total assets, we can solve for net fixed assets. Using the numbers given for the current ratio and the current liabilities, we solve for current assets:

Current ratio = Current assets/Current liabilities
Current assets = Current ratio(Current liabilities)
Current assets = 1.35($1,270)
Current assets = $1,714.50

To find the total assets, we must first find the total debt and equity from the information given. So, we find the net income using the profit margin:

Profit margin = Net income/Sales
Net income = Profit margin(Sales)
Net income = .083($8,940)
Net income = $742.02

We now use the net income figure as an input into ROE to find the total equity:

ROE = Net income/Total equity
Total equity = Net income/ROE
Total equity = $742.02/.132
Total equity = $5,621.36

Next, we need to find the long-term debt. The long-term debt ratio is:

Long-term debt ratio = .45 = LTD/(LTD + Total equity)

Inverting both sides gives:

1/.45 = (LTD + Total equity)/LTD = 1 + (Total equity/LTD)

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Substituting the total equity into the equation and solving for long-term debt gives the following:

2.222 = 1 + ($5,621.36/LTD)
LTD = $5,621.36/1.222
LTD = $4,599.30

Now, we can find the total debt of the company:

Total debt = Current liabilities + LTD
Total debt = $1,270 + 4,599.30
Total debt = $5,869.30

And, with the total debt, we can find the TD&E, which is equal to TA:

Total assets = Total debt + Total equity
Total assets = $5,869.30 + 5,621.36
Total assets = $11,490.66

And finally, we are ready to solve the balance sheet identity as:

Net fixed assets = Total assets – Current assets
Net fixed assets = $11,490.66 – 1,714.50
Net fixed assets = $9,776.16

29. Child: Profit margin = Net income/Sales

Profit margin = $1.50/$50
Profit margin = .030, or 3.0%

Store: Profit margin = Net income/Sales
Profit margin = $10,650,000/$710,000,000
Profit margin = .0150, or 1.50%

The advertisement is referring to the store's profit margin, but a more appropriate earnings measure for the firm's owners is the return on equity.

ROE = NI/TE = NI/(TA – TD)
ROE = $10,650,000/($390,000,000 – 280,000,000)
ROE = .0968, or 9.68%

30. To calculate the profit margin, we first need to calculate the sales. Using the days' sales in receivables, we find the receivables turnover is:

Days' sales in receivables = 365 days/Receivables turnover
27.53 days = 365 days/Receivables turnover
Receivables turnover = 13.26 times

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Now, we can use the receivables turnover to calculate the sales as:

Receivables turnover = Sales/Receivables
13.26 = Sales/$145,360
Sales = $1,927,221

So, the profit margin is:

Profit margin = Net income/Sales
Profit margin = $130,571/$1,927,221
Profit margin = .0678, or 6.78%

The total asset turnover is:

Total asset turnover = Sales/Total assets
Total asset turnover = $1,927,221/$785,160
Total asset turnover = 2.45 times

We need to use the DuPont identity to calculate the return on equity. Using this relationship, we get:

ROE = (Profit margin)(Total asset turnover)(1 + Debt-equity ratio)
ROE = (.0678)(2.45)(1 + .25)
ROE = .2079, or 20.79%

31. This problem requires us to work backward through the income statement. First, recognize that Net income = $(1 - T_C)$EBT. Plugging in the numbers given and solving for EBT, we get:

EBT = $18,451/(1 – .21)
EBT = $23,355.70

Now, we can add interest to EBT to get EBIT as follows:

EBIT = EBT + Interest
EBIT = $23,355.70 + 4,131
EBIT = $27,486.70

To get EBITD (earnings before interest, taxes, and depreciation), the numerator in the cash coverage ratio, add depreciation to EBIT:

EBITD = EBIT + Depreciation
EBITD = $27,486.70 + 5,193
EBITD = $32,679.70

Now, we can plug the numbers into the cash coverage ratio and calculate:

Cash coverage ratio = EBITD/Interest
Cash coverage ratio = $32,679.70/$4,131
Cash coverage ratio = 7.91 times

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32. To find the times interest earned, we need the EBIT and interest expense. EBIT is sales minus costs minus depreciation, so:

EBIT = Sales – Costs – Depreciation
EBIT = $613,000 – 245,380 – 62,900
EBIT = $304,720

Now, we need the interest expense. We know the EBIT, so if we find the taxable income (EBT), the difference between these two is the interest expense. To find EBT, we must work backward through the income statement. We need total dividends paid. We can use the dividends per share equation to find the total dividends. Doing so, we find:

DPS = Dividends/Shares
$1.51 = Dividends/20,000
Dividends = $30,200

Net income is the sum of dividends and addition to retained earnings, so:

Net income = Dividends + Addition to retained earnings
Net income = $30,200 + 70,600
Net income = $100,800

We know that the taxes are the taxable income times the tax rate. The net income is the taxable income minus taxes. Rearranging this equation, we get:

Net income = EBT – T_C(EBT)
Net income = (1 – T_C)(EBT)
$100,800 = (1 – .22)(EBT)
EBT = $129,231

Now, we can use the income statement relationship:

EBT = EBIT – Interest
$129,231 = $304,720 – Interest
Interest = $175,489

So, the times interest earned ratio is:

Times interest earned = EBIT/Interest
Times interest earned = $304,720/$175,489
Times interest earned = 1.74 times

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33. To find the return on equity, we need the net income and total equity. We can use the total debt ratio to find the total assets as:

Total debt ratio = Total debt/Total assets
.43 = $691,000/Total assets
Total assets = $1,606,977

Using the balance sheet relationship that total assets is equal to total liabilities and equity, we find the total equity is:

Total assets = Total debt + Equity
$1,606,977 = $691,000 + Equity
Equity = $915,977

We have the return on equity and the equity. We can use the return on equity equation to find net income is:

ROE = Net income/Equity
.1180 = Net income/$915,977
Net income = $108,085

We have all the information necessary to calculate the ROA. Doing so, we find the ROA is:

ROA = Net income/Total assets
ROA = $108,085/$1,606,977
ROA = .0673, or 6.73%

34. Profit margin = Net income/Sales
Profit margin = –£24,387/£305,650
Profit margin = –.0798, or –7.98%

As long as both net income and sales are measured in the same currency, there is no problem; in fact, except for some market value ratios like EPS and BVPS, none of the financial ratios discussed in the text are measured in terms of currency. This is one reason why financial ratio analysis is widely used in international finance to compare the business operations of firms and/or divisions across national economic borders. The net income in dollars is:

Net income = Profit margin × Sales
Net income = –.0798($365,412)
Net income = –$29,155.25

35. *Short-term solvency ratios:*

Current ratio	= Current assets/Current liabilities
Current ratio$_{2024}$	= $68,074/$61,722 = 1.10 times
Current ratio$_{2025}$	= $79,974/$71,677 = 1.12 times

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Quick ratio = (Current assets – Inventory)/Current liabilities
Quick ratio$_{2024}$ = ($68,074 – 27,931)/$61,722 = .65 times
Quick ratio$_{2025}$ = ($79,974 – 32,586)/$71,677 = .66 times

Cash ratio = Cash/Current liabilities
Cash ratio$_{2024}$ = $26,450/$61,722 = .43 times
Cash ratio$_{2025}$ = $29,106/$71,677 = .41 times

Asset utilization ratios:
Total asset turnover = Sales/Total assets
Total asset turnover = $422,045/$478,319 = .88 times

Inventory turnover = Cost of goods sold/Inventory
Inventory turnover = $291,090/$32,586 = 8.93 times

Receivables turnover = Sales/Accounts receivable
Receivables turnover = $422,045/$18,282 = 23.09 times

Long-term solvency ratios:
Total debt ratio = (Total assets – Total equity)/Total assets
Total debt ratio$_{2024}$ = ($425,239 – 268,517)/$425,239 = .37 times
Total debt ratio$_{2025}$ = ($478,319 – 306,642)/$478,319 = .36 times

Debt-equity ratio = Total debt/Total equity
Debt-equity ratio$_{2024}$ = ($61,722 + 95,000)/$268,517 = .58 times
Debt-equity ratio$_{2025}$ = ($71,677 + 100,000)/$306,642 = .56 times

Equity multiplier = 1 + D/E
Equity multiplier$_{2024}$ = 1 + .58 = 1.58 times
Equity multiplier$_{2025}$ = 1 + .56 = 1.56 times

Times interest earned = EBIT/Interest
Times interest earned = $93,900/$16,400 = 5.73 times

Cash coverage ratio = (EBIT + Depreciation)/Interest
Cash coverage ratio = ($93,900 + 37,055)/$16,400 = 7.99 times

Profitability ratios:
Profit margin = Net income/Sales
Profit margin = $58,125/$422,045 = .1377, or 13.77%

Return on assets = Net income/Total assets
Return on assets = $58,125/$478,319 = .1215, or 12.15%

Return on equity = Net income/Total equity
Return on equity = $58,125/$306,642 = .1896, or 18.96%

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36. The DuPont identity is:

ROE = (PM)(TAT)(EM)
ROE = (.1377)(.88)(1.56) = .1896, or 18.96%

37.

Earnings per share	= Net income/Shares
Earnings per share	= $58,125/15,000 = $3.88 per share
Sales per share	= Sales/Shares
Sales per share	= $422,045/15,000 = $28.14 per share
PE ratio	= Share price/Earnings per share
PE ratio	= $79/$3.88 = 20.39 times
PS ratio	= Share price/Sales per share
PS ratio	= $79/$28.14 = 2.81 times
Dividends per share	= Dividends/Shares
Dividends per share	= $20,000/15,000 = $1.33 per share
Book value per share	= Total equity/Shares
Book value per share	= $306,642/15,000 shares = $20.44 per share
Market-to-book ratio	= Share price/Book value per share
Market-to-book ratio	= $79/$20.44 = 3.86 times

38. The current ratio appears to be relatively low when compared to the median; however, it is above the lower quartile, meaning that at least 25 percent of firms in the industry have a lower current ratio. Overall, it does not appear that the current ratio is out of line with the industry. The total asset turnover is low when compared to the industry. In fact, the total asset turnover is in the lower quartile. This means that the company does not use assets as efficiently overall or that the company has newer assets than the industry. This would mean that the assets have not been depreciated, which would mean a higher book value and a lower total asset turnover. The debt-equity ratio is in line with the industry, between the median and the upper quartile. The profit margin is in the upper quartile. The company may be better at controlling costs or has a better product, which enables it to charge a premium price. It could also be negative in that the company may have too high of a margin on its sales, which could reduce overall net income.

39. To find the profit margin, we can solve the DuPont identity. First, we need to find the retention ratio. The retention ratio for the company is:

$b = 1 - .25$
$b = .75$

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Using the sustainable growth rate equation and solving for ROE, we get:

Sustainable growth rate = (ROE × *b*)/[1 – (ROE × *b*)]
.12 = [ROE(.75)]/[1 – ROE(.75)]
ROE = .1429, or 14.29%

Now we can use the DuPont identity to find the profit margin as:

ROE = PM(TAT)(EM)
.1429 = PM(1/.65)(1 + .95)
PM = (.1429)/[(1/.65)(1.95)]
PM = .0476, or 4.76%

40. The earnings per share is the net income divided by the shares outstanding. Since all numbers are in millions, the earnings per share for Pfizer was:

EPS = $5,998/5,616
EPS = $1.07

And the earnings per share for Merck was:

EPS = $862/2,538
EPS = $.34

The market-to-book ratio is the stock price divided by the book value per share. To find the book value per share, we divide the total equity by the shares outstanding. The book value per share and market-to-book ratio for Pfizer were:

Book value per share = $95,661/5,616
Book value per share = $17.03

Market-to-book = $27.94/$17.03
Market-to-book = 1.64 times

And the market-to-book ratio for Merck was:

Book value per share = $45,911/2,538
Book value per share = $18.09

Market-to-book = $108.33/$18.09
Market-to-book = 5.99 times

The price-earnings ratio for Pfizer was:

PE = $27.94/$1.07
PE = 26.16 times

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And for Merck, the PE was:

PE = $108.33/$.34
PE = 318.96 times

41. We are given the profit margin. Remember that:

ROA = PM(TAT)

We can calculate the ROA from the internal growth rate formula, and then use the ROA in this equation to find the total asset turnover. The retention ratio is:

b = 1 – .25
b = .75

Using the internal growth rate equation to find the ROA, we get:

Internal growth rate = (ROA × b)/[1 – (ROA × b)]
.062 = [ROA(.75)]/[1 – ROA(.75)]
ROA = .0778, or 7.78%

Plugging ROA and PM into the equation we began with and solving for TAT, we get:

ROA = (PM)(TAT)
.0778 = .073(TAT)
TAT = .0778/.073
TAT = 1.07 times

42. We should begin by calculating the D/E ratio. We calculate the D/E ratio as follows:

Total debt ratio = .45 = TD/TA

Inverting both sides we get:

1/.45 = TA/TD

Next, we need to recognize that:

TA/TD = 1 + TE/TD

Substituting this into the previous equation, we get:

1/.45 = 1 + TE/TD

Subtracting 1 (one) from both sides and inverting again, we get:

D/E = 1/[(1/.45) – 1]
D/E = .82

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With the D/E ratio, we can calculate the EM and solve for ROE using the DuPont identity:

ROE = (PM)(TAT)(EM)
ROE = (.059)(1.35)(1 + .82)
ROE = .1448, or 14.48%

Now we can calculate the retention ratio as:

b = 1 – .40
b = .60

Finally, putting all the numbers we have calculated into the sustainable growth rate equation, we get:

Sustainable growth rate = (ROE × b)/[1 – (ROE × b)]
Sustainable growth rate = [.1448(.60)]/[1 – .1448(.60)]
Sustainable growth rate = .0952, or 9.52%

43. To calculate the sustainable growth rate, we first must calculate the retention ratio and ROE. The retention ratio is:

b = 1 – $7,500/$16,300
b = .5399

And the ROE is:

ROE = $16,300/$77,000
ROE = .2117, or 21.17%

So, the sustainable growth rate is:

Sustainable growth rate = (ROE × b)/[1 – (ROE × b)]
Sustainable growth rate = [.2117(.5399)]/[1 – .2117(.5399)]
Sustainable growth rate = .1290, or 12.90%

If the company grows at the sustainable growth rate, the new level of total assets is:

New TA = 1.1290($67,000 + 77,000) = $162,580.65

To find the new level of debt in the company's balance sheet, we take the percentage of debt in the capital structure times the new level of total assets. The additional borrowing will be the new level of debt minus the current level of debt. So:

New TD = [D/(D + E)](TA)
New TD = [$67,000/($67,000 + 77,000)]($162,580.65)
New TD = $75,645.16

And the additional borrowing will be:

Additional borrowing = $75,645.16 – 67,000

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Additional borrowing = $8,645.16

The growth rate that can be supported with no outside financing is the internal growth rate. To calculate the internal growth rate, we first need the ROA, which is:

ROA = $16,300/($67,000 + 77,000)
ROA = .1132, or 11.32%

This means the internal growth rate is:

Internal growth rate = (ROA × *b*)/[1 – (ROA × *b*)]
Internal growth rate = [.1132(.5399)]/[1 – .1132(.5399)]
Internal growth rate = .0651, or 6.51%

44. We need the ROE to calculate the sustainable growth rate. The ROE is:

ROE = (PM)(TAT)(EM)
ROE = (.064)(1/1.05)(1 + .65)
ROE = .1006, or 10.06%

Now we can use the sustainable growth rate equation to find the retention ratio as:

Sustainable growth rate = (ROE × *b*)/[1 – (ROE × *b*)]
Sustainable growth rate = .12 = [.1006(*b*)]/[1 – .1006(*b*)]
b = 1.07

This implies the payout ratio is:

Payout ratio = 1 – *b*
Payout ratio = 1 – 1.07
Payout ratio = –.07

This answer indicates a dividend payout ratio of negative 7 percent, which is impossible. So, the growth rate is not consistent with the other constraints. The lowest possible payout rate is 0 (without issuing stock), which corresponds to a retention ratio of 1, or total earnings retention. This problem illustrates a key point we made in the chapter: Sustainable growth analysis forces the user to make internally consistent assumptions.

As an aside, we should note that it is possible to have a retention ratio greater than 1 if the company issues new stock. However, since the growth rate we are evaluating is perpetual, the company would have to issue stock every year, forever. But, doing so violates our underlying assumption that the sustainable growth rate requires no new equity.

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The maximum sustainable growth rate for this company is:

Maximum sustainable growth rate = (ROE × *b*)/[1 – (ROE × *b*)]
Maximum sustainable growth rate = [.1006(1)]/[1 – .1006(1)]
Maximum sustainable growth rate = .1118, or 11.18%

45. Using the beginning of period total assets, the ROA is:

ROA_{Beg} = $1,241/$15,803
ROA_{Beg} = .0785, or 7.85%

Using the end of period total assets, the ROA is:

ROA_{End} = $1,241/$16,243
ROA_{End} = .0764, or 7.64%

The ROE using beginning of period equity is:

ROE_{Beg} = $1,241/$2,795
ROE_{Beg} = .4440, or 44.40%

The ROE using the end of period equity is:

ROE_{End} = $1,241/$3,235
ROE_{End} = .3836, or 38.36%

The retention ratio, which is one minus the dividend payout ratio, is:

b = 1 – Dividends/Net income
b = 1 – $801/$1,241
b = .3546, or 35.46%

With the growth rate equations, we need to use the ROA and ROE based on the end of period assets or equity, so the internal growth rate is:

Internal growth rate = [(ROA)(*b*)]/[1 – (ROA)(*b*)]
Internal growth rate = [(.0764)(.3546)]/[1 – (.0764)(.3546)]
Internal growth rate = .0278, or 2.78%

And the sustainable growth rate is:

Sustainable growth rate = [(ROE)(*b*)]/[1 – (ROE)(*b*)]
Sustainable growth rate = [(.3836)(.3546)]/[1 – (.3836)(.3546)]
Sustainable growth rate = .1574 or 15.74%

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Using ROA × *b* and end of period assets to find the internal growth rate, we find:

Internal growth rate = $ROA_{End} \times b$
Internal growth rate = .0764 × .3546
Internal growth rate = .0271, or 2.71%

And, using ROE × *b* and the end of period equity to find the sustainable growth rate, we find:

Sustainable growth rate = $ROE_{End} \times b$
Sustainable growth rate = .3836 × .3546
Sustainable growth rate = .1360, or 13.60%

Using ROA × *b* and beginning of period assets to find the internal growth rate, we find:

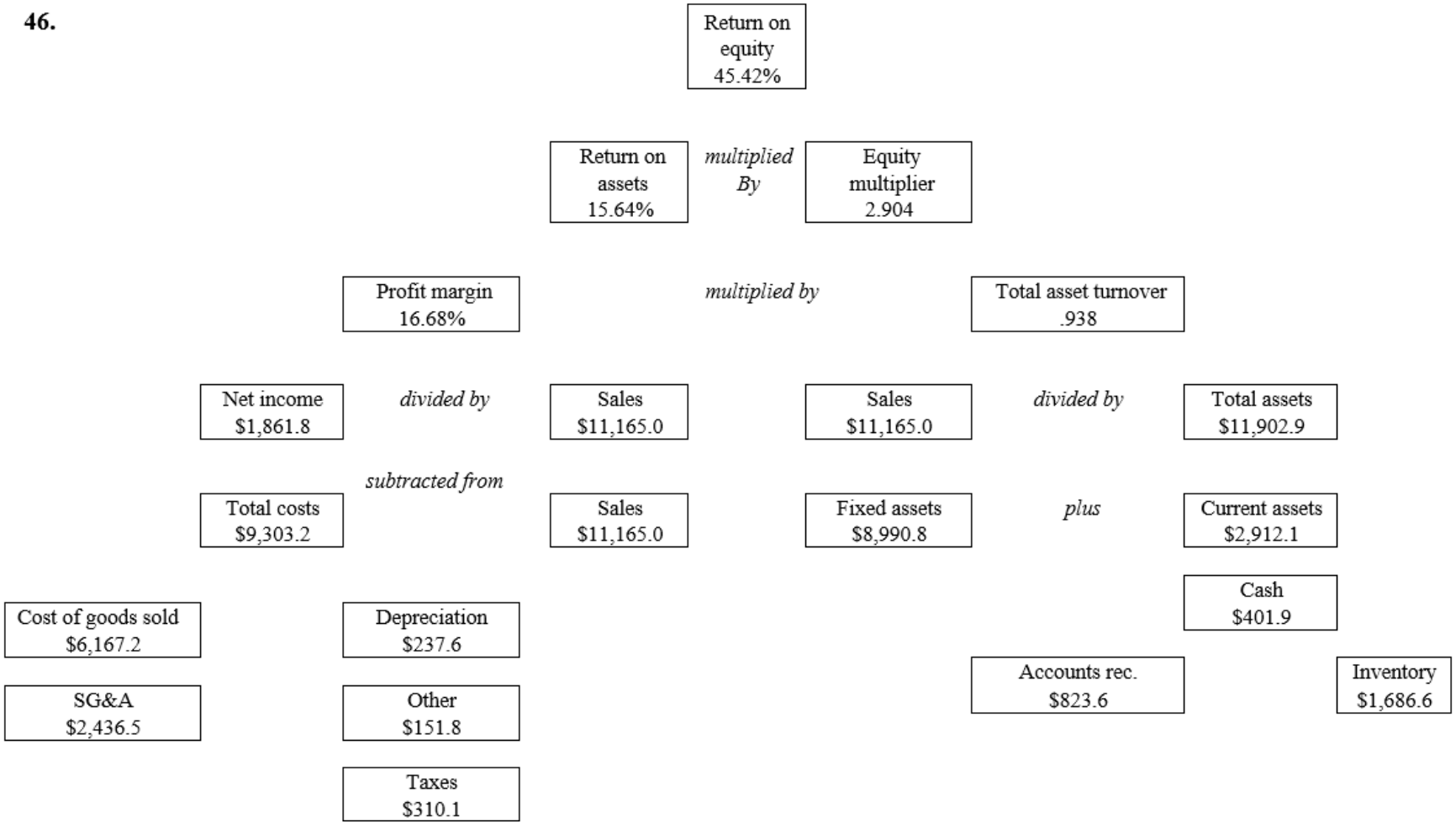
Internal growth rate = $ROA_{Beg} \times b$
Internal growth rate = .0785 × .3546
Internal growth rate = .0278, or 2.78%

And, using ROE × *b* and the beginning of period equity to find the sustainable growth rate, we find:

Sustainable growth rate = $ROE_{Beg} \times b$
Sustainable growth rate = .4440 × .3546
Sustainable growth rate = .1574, or 15.74%

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46.



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